Noha Khaled

Analytics Engineer · Mechanical Engineering & Team Leadership Background

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Technical Skills

- Data Engineering & Warehousing:
  (dbt core, DuckDB, Snowflake, Prefect)
- Database Design
  (Relational schema, surrogate keys, intentional de-normalization)
- SQL/MySQL
  (CTEs, window fn., multi-table joins, query optimization)
- Python
  (Pandas, NumPy, SQLAlchemy, fuzzy matching, regex-based parsing, automation and verification scripting)
- Git
  (Branching, pull requests, repository organization)
- BI & Reporting
  (Power BI, Excel, Power Query, DAX, Data Modelling)
- Statistics & Analysis:
  (Descriptive statistics, probability, sampling distributions, confidence intervals, hypothesis testing, correlation)
- Business Analysis & Agile:
  (Requirements, user stories, BPMN, Scrum/Kanban, prioritization, traceability)

Work Experience

Lead Technical Analyst — Optumatics Jan 2023 – June 2026

- **Engineering Simulation Data Pipeline** *(initiated and built independently within the role)*
  - Designed a normalized schema from scratch consolidating simulation data scattered across multiple sources (JSON profile files, Azure boards, reference CSVs) into a single query-able source
  - Built a Python ETL pipeline (pandas, SQLAlchemy, regex, JSON) to extract, clean, and load the multi-source data into MySQL in correct parent-child order
  - Engineered a two-step fuzzy-matching solution (python library: thefuzz) with confidence thresholds and manual-review export to link records across sources
  - Implemented duplicate and change detection logic (through mesh-file comparison) to keep imports unrepeated across repeated team runs
  - Extended independently post company closure, migrating the system to Azure Database for MySQL as an ongoing portfolio project
  - Case study: nohakhaled.com/work/case-engineering-pipeline/
- **Team Management:** Train and lead a team of 15 junior engineers; conduct regular evaluations; implement processes to improve efficiency and autonomy.
- **Client Communication (Stellantis N.V.):** Handle work requests, negotiate deliverables, and present simulation findings with design recommendations.

Technical Analyst — Optumatics Oct 2018 – Dec 2022

- Thermal and fluid modelling of vehicle parts; validated simulation results against real-world data.
- Design research and stakeholder presentations on high-voltage lithium battery packs, focusing on structural integrity and thermal runaway events.

Projects

Analysis of Olist Brazilian Ecommerce Dataset (MySQL + dbt Pipeline) June 2026

- Setup a MySQL relational database of 9 tables for 90K+ orders.
- Conducted end-to-end in SQL (CTEs, window functions, multi-table joins) and reported findings in Power BI, surfacing data-quality discrepancies and business insights across payments, delivery, and pricing.
- Rebuilt the pipeline in dbt Core + DuckDB with a full star schema (staging - intermediate - mart), correcting a grain error that had inflated customer-retention figures in the original version.
- Case study: nohakhaled.com/work/case-olist-dbt/

Certificates

Google Data Analytics Professional Certificate — Coursera April 2026

- Completed 8-course series covering the full data analysis process: Ask, Prepare, Process, Analyze, Share, and Act.

Education

Bachelor of Science in Mechanical Engineering — Cairo University, Faculty of Engineering
Graduation Project: Designing, modelling, and testing the aerodynamic components of a Formula Student Racing Car — Jul 2018